

CITYTWIN · PROBLEM STATEMENT

CityTwin problem statement

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-01	Amish Chadha	Kiosk parts cost updated for the constructable design (CTW-DDR-003)
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update per CTW-DDR-001 and CTW-CAL-001 (reference neighborhood rates, pilot siting, budget scope, adopted license and privacy settings, sibling upload paths)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, prior work, open questions)
0.1	2026-09-25	Amish Chadha	Initial scaffold

CityTwin problem statement

City data sits in separate vendor dashboards, and residents rarely see the data collected on their streets. Where street sensing has been proposed without clear rules on who holds the data and what leaves the device, it has lost public trust.

■ The problem

A city that installs counters, air and noise sensors, flood gauges and curb sensors usually buys each from a different vendor, with a different dashboard, data format and contract. The data stays in those silos, so a planner cannot put pedestrian counts next to air quality or flooding on one map, and a resident walking past a sensor cannot see what it measures, who is responsible for it, or the data it has collected.

Trust is the second problem. In a 2019 survey, 66 % of U.S. adults said the potential risks of government data collection about them outweigh the benefits, and 84 % said they have very little or no control over the data government collects about them ([Pew Research Center, 2019](#)). In Toronto, Ontario's former privacy commissioner Ann Cavoukian resigned from the Sidewalk Labs project because de-identification at the source of collection could not be guaranteed for all participants ([CBC News](#)); the project was later withdrawn on May 7, 2020, with Sidewalk Labs citing economic uncertainty ([Sidewalk Labs](#)).

The lab's smart city set (CurbCount, CrossSafe, PotholeLog, DockHub, LoadZone, HeatMap Node, FloodGauge, AirStreet and NoiseMap) already follows one privacy rule: counts or levels only, with no images or audio leaving the device. What is missing is the layer that brings their data together, applies the same rules to what is published, shows it to residents in the street, and exports it as open data.

■ Prior work

- **Open sensor platforms.** Barcelona's [Sentilo](#) is an open-source sensor and actuator platform built so a city can avoid technological silos. [Civil IoT Taiwan](#) publishes air quality, water, earthquake and disaster sensor data as open data. Both are city- or nation-scale platforms, not a kit a small city or neighborhood group can run.
- **Open standards for sensor data.** The [OGC SensorThings API](#) is an open, geospatial standard for IoT observations and metadata.
- **Transparency in public space.** [Digital Trust for Places and Routines \(DTPR\)](#), stewarded by Helpful Places, is an open communication standard with icons for signage that explain what a sensor in public space does.
- **Community sensing.** [AirQo](#), a Makerere University project, runs more than 400 low-cost air quality sensors in 14 African countries and publishes the data to the public.
- **Lab dependencies.** CityTwin runs on the TwinKit gateway (LoRaWAN concentrator, MQTT broker, time-series database) and reads the nine smart city nodes listed above.

None of these combines a small open gateway, one privacy rule set across many node types, a public street kiosk and an open data export in one design that a small team can build and audit.

■ Users and context

Table 1. Users and needs.

User	Need	Context
Resident or passer-by	See what is measured on their street, by whom, and the current values, without an app	At the kiosk on a sidewalk, square or transit stop; on a phone through the QR code
City planner or engineer	One map of counts, curb use, air, noise, heat and flooding to compare before and after a street change	Office, on the city network
Data protection officer	Evidence that published data holds no personal data and follows retention rules	Privacy impact assessment and audit
Community group or school	Open files to analyze, and a kiosk to host at a library or community center	Local projects and teaching
Researcher or journalist	Documented, reusable open data under an open license	Download or API
Maintenance crew	Node health and alerts (battery, last seen, flood alerts)	Depot or field

Operating environment (kiosk). Outdoors on a sidewalk or plaza, in rain, sun and dust, from about -20 °C to +45 °C air temperature depending on the city, with exposure to vandalism. Mains power from a street lighting circuit or an adjacent building. Indoor variants (library, city hall lobby) are simpler. Under CTW-DDR-001 D3 (decided by Amish, 2026-09-25) the pilot kiosk stands on a sheltered site with no direct sun on the screen, and under CTW-DDR-002 its air range is 0 to 35 °C, because the color e-paper is rated only 0 to 40 °C and needs margin; the full street range is the target after the pilot (CTW-REQ-001 R20). Any outdoor site gets the cabinet sun shield and a shaded screen orientation as site rules, and a panel heater only if the partner city has frost.

Reference neighborhood (for sizing, CTW-DDR-001 D9). 46 LoRaWAN nodes: 8 CurbCount, 4 CrossSafe, 12 LoadZone pucks, 6 HeatMap Node, 4 FloodGauge, 6 AirStreet and 6 NoiseMap, plus PotholeLog on 3 buses and one DockHub. With the sibling repos' TRL 3 reporting rates they send 5,616 uplinks a day, within the 50 nodes and 10,000 records a day that one TwinKit gateway must carry (CTW-CAL-001). PotholeLog uploads its daily summaries to the fleet operator's server over depot Wi-Fi, and DockHub syncs over LTE-M to its own back end; CityTwin pulls their published aggregates by outbound connections (D11, decided by Amish on 2026-09-25; formats still to be agreed with those repos).

■ Constraints

- Garage-buildable prototype. `budget_usd` is \$750 (raised from \$600 under CTW-DDR-002) and, under D1, covers the pilot kiosk only; the TwinKit gateway is costed in TwinKit. The pilot kiosk parts cost \$778.00 after the parts added to make it buildable (CTW-DDR-003), \$28.00 over `budget_usd` ; the outdoor sun shield adds \$30.00.
- Privacy by design: only counts and levels are taken in, stored or published; no images, audio or personal identifiers.
- Open software (MIT), open hardware (CERN-OHL-S-2.0) and open data under CC BY 4.0 (D7).
- Runs offline on one TwinKit gateway; no dependency on a vendor cloud.
- Mains work by a qualified electrician; street furniture installed only with the asset owner's permission.

■ Out of scope

- Any camera, microphone, face or plate recognition, or tracking of individuals.

- Control of city systems (traffic signals, street lights). CityTwin shows and exports data; it does not actuate.
- Emergency alerting. FloodGauge alerts appear on the map, but CityTwin is not a public warning system.
- Large touch-screen advertising kiosks.

■ Open questions

- Which city or neighborhood partner hosts the first kiosk? Proposed, awaiting Amish (CTW-DDR-001 01).
- Open data license: CC BY 4.0 (D7), decided by Amish on 2026-09-25.
- Small-count threshold of 5 per hour and two years of raw data retention (D6), decided by Amish on 2026-09-25.
- Will PotholeLog's operators and DockHub publish aggregates that CityTwin can pull, and will CrossSafe add a LoRaWAN uplink? To agree with those projects.

■ User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [] Run co-design sessions with intended users; record who, where and what was learned
- [] Validate load, distance, terrain and cost assumptions in the field
- [] Revise requirements (REQ) from findings before freezing the design